

1 Contest

2 Data structures

3 Strings

4 Bitwise

Contest (1)

template.cpp28 lines

```
#include <bits/stdc++.h>
#include <ext/pb_ds/assoc_container.hpp> // for PBDS
#include <ext/pb_ds/tree_policy.hpp> // for PBDS
using namespace __gnu_pbds; // for PBDS
using namespace std;
#define int long long int
const int MOD = 1000000007;

template<typename T>
using ordered_set = tree<T, null_type, less_equal<T>, rb_tree_tag,
    tree_order_statistics_node_update>;
// less_equal will make it multiset
// less makes it set
void solve() {

}

signed main() {
ios_base::sync_with_stdio(false);
cin.tie(NULL);
ios_base::sync_with_stdio(0), cin.tie(0), cout.tie(0); // orz
cin.exceptions(cin.failbit); // orz
    int t = 1;
    cin >> t;
    while (t--) {
        solve();
    }
    return 0;
}
```

troubleshoot.txt52 lines

Pre-submit:
Write a few simple test cases if sample is not enough.
Are time limits close? If so, generate max cases.
Is the memory usage fine?
Could anything overflow?
Make sure to submit the right file.

Wrong answer:
Print your solution! Print debug output, as well.
Are you clearing all data structures between test cases?
Can your algorithm handle the whole range of input?
Read the full problem statement again.
Do you handle all corner cases correctly?
Have you understood the problem correctly?
Any uninitialized variables?
Any overflows?
Confusing N and M, i and j, etc.?
Are you sure your algorithm works?
What special cases have you not thought of?
Are you sure the STL functions you use work as you think?
Add some assertions, maybe resubmit.
Create some testcases to run your algorithm on.

1 Go through the algorithm for a simple case.
Go through this list again.
1 Explain your algorithm to a teammate.
Ask the teammate to look at your code.
Go for a small walk, e.g. to the toilet.
3 Is your output format correct? (including whitespace)
Rewrite your solution from the start or let a teammate do it.

4 Runtime error:
Have you tested all corner cases locally?
Any uninitialized variables?
Are you reading or writing outside the range of any vector?
Any assertions that might fail?
Any possible division by 0? (mod 0 for example)
Any possible infinite recursion?
Invalidated pointers or iterators?
Are you using too much memory?
Debug with resubmits (e.g. remapped signals, see Various).

Time limit exceeded:
Do you have any possible infinite loops?
What is the complexity of your algorithm?
Are you copying a lot of unnecessary data? (References)
How big is the input and output? (consider scanf)
Avoid vector, map. (use arrays/unordered_map)
What do your teammates think about your algorithm?

Memory limit exceeded:
What is the max amount of memory your algorithm should need?
Are you clearing all data structures between test cases?

Data structures (2)

SegmentTree.h
Description: Segment tree with ability to add or set values of large intervals, and compute max of intervals. Can be changed to other things. Use with a bump allocator for better performance, and SmallPtr or implicit indices to save memory.
Usage: Node* tr = new Node(v, 0, sz(v));
Time: $\mathcal{O}(\log N)$.

```
struct Node {
    int sum;
    Node(int sum) : sum(sum) {}
    Node() : sum(0) {}
};
```

```
Node merge(Node a,Node b) {
    return Node(a.sum + b.sum);
}
```

```
void build(int idx,int l,int r,vector<int> &a) {
    if(l==r) {
        tree[idx] = {a[l]};
        return;
    }
    int mid = (l+r)/2;
    build(2*idx,l,mid,a);
    build(2*idx + 1,mid + 1,r,a);
    tree[idx] = merge(tree[2*idx],tree[(2*idx)+ 1]);
}
```

```
Node query(int idx,int tl,int th,int ql,int qh) {
    if(ql>th || qh<tl) {
        return Node();
    }
    if(ql<=tl && th<=qh) {
        return tree[idx];
    }
```

```
}
    int mid = (tl+th)/2;
    return merge(query(2*idx,tl,mid,ql,qh),query(2*idx + 1,mid
        + 1,th,ql,qh));
}

    if(tl>pos || th<pos) {
        return;
    }
    if(tl==th) {
        tree[idx] = {val};
        return;
    }
    int mid = (tl+th)/2;
    update(2*idx,tl,mid,pos,val);
    update(2*idx + 1,mid + 1,th,pos,val);
    tree[idx] = merge(tree[2*idx],tree[2*idx + 1]);
}
```

```
// you can change the struct to whatever you want
tree.resize(4*N);
build(1,1,N,a);
update(1,1,N,pos,x);
query(1,1,N,l,r).sum;
```

LazySegmentTree.h
Description: Segment tree with ability to add or set values of large intervals, and compute max of intervals. Can be changed to other things. Use with a bump allocator for better performance, and SmallPtr or implicit indices to save memory.
Usage: Node* tr = new Node(v, 0, sz(v));
Time: $\mathcal{O}(\log N)$.

```
struct Node {
    int sum;
    int lazy_add;
    int lazy_set;
    Node(int sum,int lazy_add,int lazy_set) : sum(sum),lazy_add
        (lazy_add),lazy_set(lazy_set) {}
    Node() : sum(0),lazy_add(0),lazy_set{-1} {}
};
```

```
Node merge(Node a,Node b) {
    return Node(a.sum+b.sum,0,-1);
}
```

```
vector<Node> tree;
```

```
void apply_add(int idx,int l,int r,int val) {
    tree[idx].sum += (r-l+1)*val;
    if(tree[idx].lazy_set!=-1) {
        tree[idx].lazy_set += val;
    } else {
        tree[idx].lazy_add += val;
    }
}
```

```
void apply_set(int idx,int l,int r,int val) {
    tree[idx].sum = (r-l+1)*val;
    tree[idx].lazy_set = val;
    tree[idx].lazy_add = 0;
}
```

```
void build(int idx,int l,int r,vector<int> &a) {
    if(l==r) {
        tree[idx] = {a[l],0,-1};
        return;
    }
    int mid = (l+r)/2;
```

```
    build(2*idx,l,mid,a);
    build(2*idx + 1,mid + 1,r,a);
    tree[idx] = merge(tree[2*idx],tree[(2*idx) + 1]);
}

void push(int idx,int l,int r) {
    if(l==r) {
        return;
    }
    int mid = (l+r)/2;
    if(tree[idx].lazy_set!=-1) {
        apply_set(2*idx,l,mid,tree[idx].lazy_set);
        apply_set(2*idx + 1,mid + 1,r,tree[idx].lazy_set);
        tree[idx].lazy_set = -1;
    }
    if(tree[idx].lazy_add!=0) {
        apply_add(2*idx,l,mid,tree[idx].lazy_add);
        apply_add(2*idx + 1,mid + 1,r,tree[idx].lazy_add);
        tree[idx].lazy_add = 0;
    }
}

Node query(int idx,int tl,int th,int ql,int qh) {
    if(ql<=tl && th<=qh) {
        return tree[idx];
    }
    if(ql>th || qh<tl) {
        return Node();
    }
    int mid = (tl+th)/2;
    push(idx,tl,th);
    return merge(query(2*idx,tl,mid,ql,qh),query(2*idx + 1,mid
+1,th,ql,qh));
}

void update(int idx,int tl,int th,int ql,int qh,int val,int
type) {
    if(ql>th || qh<tl) {
        return;
    }

    if(ql<=tl && th<=qh) {
        if(type==1) {
            // assign
            apply_set(idx,tl,th,val);
        } else if(type==2) {
            apply_add(idx,tl,th,val);
            // add
        }
        return;
    }
    int mid = (tl+th)/2;
    push(idx,tl,th);
    update(2*idx,tl,mid,ql,qh,val,type);
    update(2*idx + 1,mid + 1,th,ql,qh,val,type);
    tree[idx] = merge(tree[2*idx],tree[2*idx + 1]);
}

// // Usage :

// tree.resize(4*N);
// build(1,1,N,a);
// update(1,1,N,l,r,v,1);
// update(1,1,N,l,r,v,2);
// query(1,1,N,l,r).sum;
```

FenwickTree.h

Description: Computes partial sums $a[0] + a[1] + \dots + a[pos - 1]$, and updates single elements $a[i]$, taking the difference between the old and new value.

Time: Both operations are $\mathcal{O}(\log N)$.

f2c389, 18 lines

```
struct BIT {
    int size;
    vector<int> bit;

    BIT(int n) : size(n),bit(n+1){}

    void update(int x, int v) {
        x++;
        for (; x <= size; x += x & (-x)) { bit[x] += v; }
    }
    int query(int b) {
        b++;
        int result = 0;
        for (; b > 0; b -= b & (-b)) { result += bit[b]; }
        return result;
    }
};
```

FenwickTree2d.h

Description: Computes sums $a[i,j]$ for all $i < I, j < J$, and increases single elements $a[i,j]$. Requires that the elements to be updated are known in advance (call fakeUpdate() before init()).

Time: $\mathcal{O}(\log^2 N)$. (Use persistent segment trees for $\mathcal{O}(\log N)$.)

fd19a8, 24 lines

```
struct BIT {
    int n,m;
    vector<vector<int>> bit;

    BIT(int n,int m) : n(n),m(m),bit(n+1,vector<int> (m+1)){}

    void update(int r, int c,int v) {
        r++;
        c++;
        for (; r <= n; r += r & -r) {
            for (int i = c; i <= m; i += i & -i) { bit[r][i] +=
                v; }
        }
    }

    int query(int r,int c) {
        r++;
        c++;
        int sum = 0;
        for (; r > 0; r -= r & -r) {
            for (int i = c; i > 0; i -= i & -i) { sum += bit[r
                ][i]; }
        }
        return sum;
    }
};
```

sqrtDecomposition.h

Description: SQRT Decomposition

3decba, 40 lines

```
struct Sqrt {
    int block_size;
    vector<int> nums;
    vector<long long> blocks;
    Sqrt(int sqtrn, vector<int> &arr) : block_size(sqtrn), blocks
        (sqtrn, 0) {
        nums = arr;
        for (int i = 0; i < nums.size(); i++) { blocks[i /
            block_size] += nums[i]; }
    }
};
```

```
    }
    void update(int x, int v) {
        blocks[x / block_size] -= nums[x];
        nums[x] = v;
        blocks[x / block_size] += nums[x];
    }
    long long query(int r) {
        long long res = 0;
        for (int i = 0; i < r / block_size; i++) { res += blocks[i
            ]; }
        for (int i = (r / block_size) * block_size; i < r; i++) {
            res += nums[i]; }
        return res;
    }
    long long query(int l, int r) { return query(r) - query(l -
        1); }
};

int main() {
    int n, q;
    cin >> n >> q;

    vector<int> arr(n);
    for (int i = 0; i < n; i++) { cin >> arr[i]; }
    Sqrt sq((int)ceil(sqrt(n)), arr);

    for (int i = 0; i < q; i++) {
        int t, l, r;
        cin >> t >> l >> r;
        if (t == 1) {
            sq.update(l - 1, r);
        } else {
            cout << sq.query(l, r) << "\n";
        }
    }
}
```

binaryLifting.h

Description: Useful for LCA and querying on paths of the tree

9b4453, 41 lines

```
const int N = 200005;
const int LOG = 22;
int parent[N+1][LOG+1];
void dfs(int u,int par,vector<vector<int>> &g,vector<int> &
    depth) {
    parent[u][0] = par; // can add query info here
    for(auto v : g[u]) {
        if(v!=par) {
            depth[v] = depth[u] + 1;
            dfs(v,u,g,depth);
        }
    }
}

int merge(int a,int b,vector<int> &depth) {
    if(depth[a]<depth[b]) {
        swap(a,b);
    }
    int k = depth[a] - depth[b];
    for(int i=0;i<LOG;i++) {
        if(k&(1LL<<i)) {
            k = k - (1LL<<i);
            a = parent[a][i]; // can add query info here
        }
    }
    if(a==b) {
        return a;
    }
    for(int i=LOG;i>=0;i--) {
        if(parent[a][i]!=parent[b][i]) {
```

```
        a = parent[a][i];
        b = parent[b][i]; // can add query info here
    }
}
return parent[a][0];
}
// Usage :
dfs(1,0,g,depth);
for(int i=1;i<=LOG;i++) {
    for(int j=1;j<=n;j++) {
        parent[j][i] = parent[parent[j][i-1]][i-1]; // here
        query also
    }
}
```

Strings (3)

doubleHasher.h
Description: Various self-explanatory methods for string hashing. Use on Codeforces, which lacks 64-bit support and where solutions can be hacked.

```
struct hasher {
    int base;
    int mod;
    vector<int> hash;
    vector<int> basepow;
    vector<int> rHash;

    hasher() {}

    hasher(string s, int _base = 37, int _mod = 1e9 + 7) {
        base = _base;
        mod = _mod;
        int n = s.size();
        s = "#" + s + "#";

        hash.resize(n + 2);
        basepow.resize(n + 2);
        rHash.resize(n + 2);

        hash[0] = 0;
        basepow[0] = 1;

        for (int i = 1; i <= n; i++) {
            hash[i] = (hash[i - 1] * base + (s[i] - 'a' + 1)) % mod;
            basepow[i] = (basepow[i - 1] * base) % mod;
        }

        rHash[n + 1] = 0;
        for (int i = n; i >= 1; i--) {
            rHash[i] = (rHash[i + 1] * base + (s[i] - 'a' + 1)) % mod;
        }
    }

    int getHash(int l, int r) {
        int res = hash[r];
        res -= (basepow[r - l + 1] * hash[l - 1]) % mod;
        res = (res % mod + mod) % mod;
        return res;
    }

    int getRevHash(int l, int r) {
        int res = rHash[l];
        res -= (basepow[r - l + 1] * rHash[r + 1]) % mod;
        res = (res % mod + mod) % mod;
        return res;
    }
}
```

```
bool isPalindrome(int l, int r) {
    return getHash(l, r) == getRevHash(l, r);
}

struct doubleHasher{
    hasher hash1,hash2;

    doubleHasher() {

    }

    doubleHasher(string s) {
        hash1 = hasher(s, 53, 1000000007);
        hash2 = hasher(s, 53, 9999999937);
    }

    int getHash(int l,int r) {
        return (1LL<<32)*hash1.getHash(l,r) + hash2.getHash(l,r);
    }
};
```

SAM.h
Description: Suffix Automaton

```
struct state{
    int len,link;
    int next[26];
    // bool is_clone; // occurence counting
    // int first_pos; // finding first position
};

const int MAXN = 1e5;
state st[2*MAXN];
int sz,last;

void sa_init() {
    st[0].len = 0;
    st[0].link = -1;
    memset(st[0].next,-1,sizeof(st[0].next));
    sz = 1;
    last = 0;
}

void sa_extend(char c) {
    int cur = sz++;
    st[cur].len = st[last].len + 1;
    // st[cur].is_clone = false;
    // st[cur].first_pos = st[cur].len - 1;
    memset(st[cur].next,-1,sizeof(st[cur].next));

    int p = last;
    int code = c - 'a';
    while(p!=-1 && st[p].next[code]==-1) {
        st[p].next[code] = cur;
        p = st[p].link;
    }
    if (p == -1) {
        st[cur].link = 0;
    } else {
        int q = st[p].next[code];
        if (st[p].len + 1 == st[q].len) {
            st[cur].link = q;
        } else {
            int clone = sz++;
            st[clone].len = st[p].len + 1;
            st[clone].link = st[q].link;
```

```
memcpy(st[clone].next, st[q].next, sizeof(st[q].next));
// st[clone].is_clone = true;
// st[clone].first_pos = st[q].first_pos;
while (p != -1 && st[p].next[code] == q) {
    st[p].next[code] = clone;
    p = st[p].link;
}
st[q].link = st[cur].link = clone;
}

last = cur;
}
```

Number of Distinct Substrings: Each state u represents substrings of lengths $(len(link(u)), len(u)]$, calculated as:

$$\sum_{u=1}^{sz-1} (len(u) - len(link(u)))$$

Total Length of Distinct Substrings: Arithmetic sum per state:

$$\sum_{u=1}^{sz-1} \frac{(len(u) - len(link(u)))(len(u) + len(link(u)) + 1)}{2}$$

Occurrence Count: Initialize $cnt[u] = 1$ for non-clones, 0 for clones; sort states by decreasing len ; propagate $cnt[link(u)] += cnt[u]$.

First Occurrence Position: Store $first_pos$: new state $= len(cur) - 1$, clone $= first_pos(q)$; query answer $first_pos(u) - |P| + 1$.

All Occurrence Positions: Store inverse links $adj[link(u)].push_back(u)$; DFS/BFS from state u on inverse link tree (ignore clones).

Longest Common Substring (2 Strings): Build SAM of S , stream T keeping node v and length l : if $next[v][c]$ exists, $v = next[v][c]$, $l + +$; else jump $v \rightarrow link(v)$, $l = len(v)$ until valid. Max l is LCS.

LCS of K Strings: Build SAM of $T = S_1\#_1S_2\#_2 \dots S_K\#_K$; propagate per-state reachability (bitmask/count); answer is max $len(u)$ reachable by all K .

Shortest Non-Appearing String: BFS/DP on SAM graph (not links), $dp[u] = 1 + \min_{(u,v) \in edges} dp[v]$; shortest path from root to state with missing transition.

Smallest Cyclic Shift: Build SAM for $S + S$, from root greedily take smallest lex edge for exactly $|S|$ steps.

K-th Lexicographical Substring: Compute DAG paths $paths[u] = 1 + \sum_{(u,v) \in edges} paths[v]$, greedily traverse from root subtracting subtree sizes until K hits.

KMP.h
Description: $pi[x]$ computes the length of the longest prefix of s that ends at x , other than $s[0...x]$ itself (abacaba -> 0010123). Can be used to find all occurrences of a string.
Time: $\mathcal{O}(n)$

```
vi pi_function(string s) {
    int n = s.size(s);
    vi pi(n, 0);
    for (int i = 1, j = 0; i < n; i++) {
```

```
        while (j > 0 && s[i] != s[j]) j = pi[j - 1];
        if (s[i] == s[j]) pi[i] = ++j;
    }
    return pi;
}

// Find occurrences of pat in text
int kmp(string text, string pat) {
    auto pi = pi_function(pat);
    int res = 0;
    for (int i = 0, j = 0; i < ssize(text); i++) {
        while (j > 0 && text[i] != pat[j]) j = pi[j - 1];
        if (text[i] == pat[j]) ++j;
        if (j == ssize(pat))
            res++; // occurrence starting at text[i-pat.size()
                  +1] found
    }
    return res;
}
```

Zfunc.h

Description: z[i] computes the length of the longest common prefix of s[i:] and s, except z[0] = 0. (abacaba -> 0010301)
Time: $\mathcal{O}(n)$

a0719e, 20 lines

```
vi z_function(string s) {
    int n = ssize(s);
    vi z(n, 0);
    for (int i = 1, l = 0, r = 0; i < n; i++) {
        if (i <= r) z[i] = min(r - i + 1, z[i - l]);
        while (i + z[i] < n && s[i + z[i]] == s[z[i]]) z[i]++;
        if (i + z[i] - 1 > r) l = i, r = i + z[i] - 1;
    }
    // z[0] = n; // add if you need z[0]=n
    return z;
}

int z_algorithm(string text, string pat) { // Find occurrences
    of pat in text
    auto z = z_function(pat + '\0' + text);
    int res = 0;
    for (int i = 0; i < ssize(text) - ssize(pat) + 1; i++)
        if (z[i + ssize(pat) + 1] == ssize(pat))
            res++; // occurrence starting at text[i] found
    return res;
}
```

Bitwise (4)

Xor Basis.h

Description: XOR Basis for Max Subset Xor and Number of Subsets
Time: Both operations are $\mathcal{O}(\log N)$.

f1749f, 34 lines

```
void solve() {
    int n;
    cin>>n;
    vector<int> a(n);
    for(int i=0;i<n;i++) {
        cin>>a[i];
    }
    sort(a.begin(),a.end());
    int index = 0;
    for(int i=32;i>=0;i--) {
        int temp = -1;
        for(int j=index;j<n;j++) {
            if(a[j]&(1LL<<i)) {
                temp = j;
                break;
            }
        }
    }
}
```

Zfunc Xor Basis Max XOR AOS

```
    }
    if(temp==-1) {
        continue;
    }
    swap(a[index],a[temp]);
    for(int j=0;j<n;j++) {
        if(a[j]&(1LL<<i) && j!=index) {
            a[j] = a[j]^a[index];
        }
    }
    index++;
}
int ans = 0;
for(int i=0;i<n;i++) {
    ans = max(ans,ans^a[i]);
}
cout<<ans<<endl;
}
```

Max XOR.h

Description: Using Tries to extract max subarray XOR
Time: Both operations are $\mathcal{O}(\log N)$.

857b00, 35 lines

```
struct trie {
    int val;
    trie* arr[2];
    trie() {
        arr[0] = NULL;
        arr[1] = NULL;
    }
};
void insert(trie *root,int n) {
    trie* temp = root;
    for(int i=32;i>=0;i--) {
        bool x = n&(1LL<<i);
        if(temp->arr[x]!=NULL) {
            temp = temp->arr[x];
        } else {
            temp->arr[x] = new trie();
            temp = temp->arr[x];
        }
    }
    temp->val = n;
}
int query(trie *root,int n) {
    trie* temp = root;
    for(int i=32;i>=0;i--) {
        bool x = n&(1LL<<i);
        if(temp->arr[!x]!=NULL) {
            temp = temp->arr[!x];
        } else if(temp->arr[x]!=NULL) {
            temp = temp->arr[x];
        }
    }
    return n^temp->val;
}
//Usage :
//trie *root = new trie();
```

AOS.h

Description: Addition and Subtraction Over Subsets
Time: Both operations are $\mathcal{O}(\log N)$.

175b47, 55 lines

```
// MLOG represents number of bits, MAXN is total masks (2^MLOG)
const lli MLOG = 20;
const lli MAXN = (1LL << MLOG);
```

```
// Forward 1: Subset Sum (SOS DP). Adds value of each element
to all its supersets.
void forward1(lli dp[]) {
    for (int bit = 0; bit < MLOG; bit++) {
        for (int i = 0; i < MAXN; i++) {
            // Check if the specific bit is set in the current
            mask i
            if (i & (1 << bit)) {
                // dp[mask] = sum of dp[submask], moving from
                smaller to larger masks
                dp[i] += dp[i ^ (1 << bit)];
            }
        }
    }
}

// Backward 1: Inverse Subset Sum. Subtracts submask values to
restore original array.
void backward1(lli dp[]) {
    for (int bit = 0; bit < MLOG; bit++) {
        // Iterate backward to reverse the prefix sum
        accumulation correctly
        for (int i = MAXN - 1; i >= 0; i--) {
            if (i & (1 << bit)) {
                // Subtract the contribution of the submask (
                where bit is 0)
                dp[i] -= dp[i ^ (1 << bit)];
            }
        }
    }
}

// Forward 2: Superset Sum. Adds value of each element to all
its subsets.
void forward2(lli dp[]) {
    for (int bit = 0; bit < MLOG; bit++) {
        for (int i = MAXN - 1; i >= 0; i--) {
            // Check if the specific bit is set in the current
            mask i
            if (i & (1 << bit)) {
                // Add the value of the supermask to the
                submask (flipping bit from 1 to 0)
                dp[i ^ (1 << bit)] += dp[i];
            }
        }
    }
}

// Backward 2: Inverse Superset Sum. The inverse operation of
forward2.
void backward2(lli dp[]) {
    for (int bit = 0; bit < MLOG; bit++) {
        // Iterate forward to reverse the superset accumulation
        correctly
        for (int i = 0; i < MAXN; i++) {
            if (i & (1 << bit)) {
                // Remove the contribution of the supermask
                from the submask
                dp[i ^ (1 << bit)] -= dp[i];
            }
        }
    }
}
```